BADASS - Defense Notes

Project Overview

What is GNS3?

GNS3 is a network simulator that allows us to build and test network topologies using virtual routers, switches, Docker containers, and virtual machines.

What is BGP?

BGP (Border Gateway Protocol) is the routing protocol used to exchange routing information between Autonomous Systems (AS). It is the protocol that powers Internet routing.

Difference Between Layer 2 and Layer 3

Layer 2	Layer 3
Uses MAC addresses	Uses IP addresses
Switching	Routing
Frames	Packets
Switches and Bridges	Routers

Part 1 - Theory

What is Packet Routing Software?

Packet routing software is software that decides where packets should be forwarded. Examples include FRRouting, Quagga, and Zebra.

What is BGPD?

BGPD is the BGP daemon responsible for establishing BGP sessions and exchanging routing information with neighbors.

What is OSPFD?

OSPFD is the OSPF daemon responsible for discovering neighbors and exchanging routes using the OSPF protocol.

What is the Routing Engine Service Requested in the Subject?

The routing engine service is IS-IS (`isisd`). It is another dynamic routing protocol used by service providers and large networks.

What is BusyBox?

BusyBox is a lightweight collection of Linux utilities packaged into a single executable. It provides commands such as `ping` , `sh` , `ifconfig` , and `ls` .

Part 2 - Theory

What is a VXLAN?

VXLAN (Virtual Extensible LAN) is an overlay technology that extends Layer 2 networks over a Layer 3 network using UDP encapsulation.

Difference Between VXLAN and VLAN

VLAN	VXLAN
12-bit ID	24-bit VNI
Maximum 4096 networks	About 16 million networks
Works in Layer 2	Works over Layer 3
Local datacenters	Large datacenters and cloud environments

What is a Switch?

A switch forwards Ethernet frames using MAC addresses and connects devices within the same Layer 2 network.

What is a Bridge?

A bridge is a software or hardware component that connects multiple Layer 2 interfaces into a single broadcast domain.

Difference Between Broadcast and Multicast

Broadcast	Multicast
Sent to everyone	Sent to a specific group
Higher bandwidth usage	More efficient

Broadcast	Multicast
One-to-all	One-to-many

Expected Topology Operation

Hosts connected to different routers communicate as if they were on the same Layer 2 network through VXLAN tunnels.

Part 2 - Static VXLAN

How Does Static VXLAN Work?

Each VTEP is manually configured with the remote VTEP IP address.

VXLAN ID

The project requires:

```
VNI = 10
```

Verification

A ping between hosts should succeed and packet inspection should show VXLAN packets encapsulated in UDP port 4789.

Part 2 - Multicast VXLAN

How Does Multicast VXLAN Work?

Instead of manually configuring remote VTEPs, VTEPs join the same multicast group and automatically discover each other.

Advantages Over Static VXLAN

- Easier to scale
- Less manual configuration
- Automatic VTEP discovery
- Better for larger environments

Part 3 - Theory

What is BGP EVPN?

BGP EVPN (Ethernet VPN) is a control plane that distributes MAC address information using BGP.

What is Route Reflection?

Route Reflection allows BGP routers to exchange routes through a central Route Reflector instead of creating a full mesh of BGP sessions.

What is a VTEP?

VTEP (VXLAN Tunnel Endpoint) is a router or switch that encapsulates and decapsulates VXLAN traffic.

What is a VNI?

VNI (VXLAN Network Identifier) identifies a VXLAN segment.

Example:

```
VNI 10
```

represents one virtual Layer 2 network.

Difference Between Type-2 and Type-3 Routes

Type-2 Route

Contains MAC address information.

Example:

```
Host MAC -> VTEP
```

Used to advertise hosts.

Type-3 Route

Contains VTEP reachability information.

Example:

```
VTEP 1.1.1.2 participates in VNI 10
```

Used for VXLAN tunnel discovery.

Expected Topology Operation

1. OSPF advertises all loopbacks.
2. BGP EVPN sessions are established with the Route Reflector.
3. Type-3 routes advertise VTEPs.
4. Hosts become active.
5. VTEPs learn host MAC addresses.
6. Type-2 routes are generated.
7. Hosts communicate through VXLAN tunnels.

Practical Validation Checklist

Part 1

- Docker images created
- Images imported into GNS3
- Routers accessible
- BGPD running
- OSPFD running
- IS-IS running
- BusyBox available

Part 2

- VXLAN ID = 10
- Bridge created
- Hosts communicate
- Packet inspection shows VXLAN traffic

Part 3

- OSPF neighbors established
- Route Reflector operational
- EVPN sessions established
- Only Type-3 routes when hosts are down
- Type-2 routes appear when hosts start
- Hosts communicate through VXLAN
- VXLAN ID = 10
- OSPF packets visible